

A Rate-Free Weighted Diagonal Bridge for Full Noisy Traces

Bin Wang

July 2026

Abstract

The archived noisy-trace/counterloop bridge was growing-prefix but unweighted. We show that the same fixed-order inputs already imply a weighted theorem after a sharper diagonal choice. At level j , choose the coefficient tolerance inversely proportional to $j \sum_{n=2}^j R^n/n$. Fixed-order noisy trace convergence and fixed-order finite-radius counterloop convergence then provide clocks $h_\sigma, k_\sigma \rightarrow \infty$, with $h_\sigma < 2k_\sigma$, on which the full weighted constituent error tends to zero. The construction remains nonquantitative: h_σ need not dominate any prescribed function of $\log(1/\sigma)$. It therefore does not reach the minimal RH-292 bridge clock and does not close the determinant interface.

1 Fixed-order inputs

Let

$$e_{\sigma,k,n} = c_{\sigma,n} - s_{k,n} - a_n.$$

For each fixed $n \geq 2$, the archived limits give

$$c_{\sigma,n} \rightarrow c_n, \quad s_{k,n} \rightarrow p_n, \quad a_n = c_n - p_n.$$

Fix $R > 1$ and define

$$A_j(R) = \sum_{n=2}^j \frac{R^n}{n}, \quad \eta_j = \frac{1}{2jA_j(R)}.$$

2 Weighted diagonal theorem

Theorem 2.1. *There are integer-valued clocks h_σ, k_σ such that*

$$h_\sigma \rightarrow \infty, \quad k_\sigma \rightarrow \infty, \quad h_\sigma < 2k_\sigma,$$

and

$$E_\sigma^{(h)}(R) := \sum_{n=2}^{h_\sigma} \frac{|e_{\sigma,k_\sigma,n}| R^n}{n} \rightarrow 0.$$

Proof. For each $j \geq 2$, fixed-order noisy convergence supplies $\delta_j > 0$ such that

$$\max_{2 \leq n \leq j} |c_{\sigma,n} - c_n| \leq \eta_j \quad (0 < \sigma \leq \delta_j).$$

Choose the δ_j strictly decreasing with $\delta_j \leq j^{-1}$. Fixed-order shell convergence supplies K_j^0 such that

$$\max_{2 \leq n \leq j} |s_{k,n} - p_n| \leq \eta_j \quad (k \geq K_j^0).$$

Set $K_1 = 0$, and recursively choose $K_j > \max\{K_{j-1}, j/2, K_j^0\}$. On the slab $\delta_{j+1} < \sigma \leq \delta_j$, set $h_\sigma = j$ and $k_\sigma = K_j$. Then $|e_{\sigma,k_\sigma,n}| \leq 2\eta_j$ for $2 \leq n \leq j$, and hence

$$E_\sigma^{(h)}(R) \leq 2\eta_j A_j(R) = j^{-1}.$$

For $\sigma > \delta_2$, define both clocks arbitrarily. Both clocks diverge and the recursive inequality gives the pre-alias condition. $\square$

3 Protocol and exact boundary

The computation lists $A_j(7/5)$, the chosen tolerance η_j , and the resulting certified budget $1/j$. It evaluates the proof schedule only; it does not infer the unknown slab thresholds δ_j from finite data.

Remark 3.1. The theorem proves a weighted full-trace constituent on an unspecified slow clock. It gives no lower bound comparable with $\log(1/\sigma)/\log(10/7)$ and says nothing about the noisy modulus head. Thus the required logarithmic E_σ and D_σ budgets remain open. Gates A–E remain false/open, with no Hilbert–Polya or RH conclusion.